

Measure-Theoretic Probability Theory

Lecture Notes for Econometrics.

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1 1. Introduction

1.1 1.1 Probability and Uncertainty

Definition 1.1. Probability is a numerical measure that quantifies uncertainty associated with events.

In economics, uncertainty is central to decision-making, from consumption choices to financial markets and policy design.

1.2 1.2 Approaches to Probability

1.2.1 Classical

Based on equally likely outcomes.

1.2.2 Frequentist

Probability as long-run relative frequency.

1.2.3 Subjective

Probability as degree of belief.

1.2.4 Axiomatic (Measure-Theoretic)

Probability is defined as a measure on a measurable space.

Remark 1.2. The axiomatic approach provides the mathematical foundation for modern econometrics and stochastic processes.

2 2. Probability Spaces

2.1 2.1 Sample Space

Definition 2.1. A sample space Ω is the set of all possible outcomes.

2.2 2.2 Sigma-Algebra

Definition 2.2. A sigma-algebra $\mathcal{F}$ is a collection of subsets of Ω such that: 1. $\Omega \in \mathcal{F}$ 2. If $A \in \mathcal{F}$, then $A^c \in \mathcal{F}$ 3. If $A_n \in \mathcal{F}$, then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{F}$

Theorem 2.3. Sigma-algebras are closed under countable intersections.

Proof. By De Morgan's laws,

$$\bigcap_{n=1}^{\infty} A_n = \left(\bigcup_{n=1}^{\infty} A_n^c \right)^c$$

and closure follows. ■

2.3 2.3 Probability Measure

Definition 2.4. A probability measure $\mathbb{P}$ is a function $\mathbb{P} : \mathcal{F} \rightarrow [0, 1]$ satisfying: 1. $\mathbb{P}(A) \geq 0$ 2. $\mathbb{P}(\Omega) = 1$ 3. $\mathbb{P}(\bigcup A_i) = \sum \mathbb{P}(A_i)$ for disjoint sets

3 3. Random Variables

3.1 3.1 Definition

Definition 3.1. A random variable is a measurable function $X : \Omega \rightarrow \mathbb{R}$.

3.2 3.2 Distribution Function

Definition 3.2. The distribution function is

$$F_X(x) = \mathbb{P}(X \leq x)$$

4 4. Distributions

4.1 4.1 CDF Properties

Theorem 4.1. Every CDF is non-decreasing, right-continuous, and satisfies:

$$\lim_{x \rightarrow -\infty} F(x) = 0, \quad \lim_{x \rightarrow \infty} F(x) = 1$$

4.2 4.2 Independence

Definition 4.2. Random variables X, Y are independent if

$$\mathbb{P}(X \in A, Y \in B) = \mathbb{P}(X \in A)\mathbb{P}(Y \in B)$$

5 5. Expectations

5.1 5.1 Expected Value

Definition 5.1. For a random variable X ,

$$\mathbb{E}[X] = \int X d\mathbb{P}$$

5.2 5.2 Properties

Theorem 5.2. Expectation is linear:

$$\mathbb{E}[aX + bY] = a\mathbb{E}[X] + b\mathbb{E}[Y]$$

Proof. Follows directly from properties of the Lebesgue integral. ■

5.3 5.3 Jensen's Inequality

Theorem 5.3. If ϕ is convex, then

$$\phi(\mathbb{E}[X]) \leq \mathbb{E}[\phi(X)]$$

6 6. Convergence Concepts

6.1 6.1 Almost Sure Convergence

Definition 6.1. $X_n \rightarrow X$ almost surely if

$$\mathbb{P}(\lim X_n = X) = 1$$

6.2 6.2 Convergence in Probability

Definition 6.2. $X_n \rightarrow X$ in probability if

$$\mathbb{P}(|X_n - X| > \varepsilon) \rightarrow 0$$

6.3 6.3 Convergence in Distribution

Definition 6.3. $X_n \rightarrow X$ in distribution if

$$F_{X_n}(x) \rightarrow F_X(x)$$

6.4 6.4 Relationships

Theorem 6.4. Almost sure convergence implies convergence in probability.

6.5 6.5 Weak Law of Large Numbers

Theorem 6.5. Let X_i be i.i.d. with finite mean μ . Then

$$\bar{X}_n \rightarrow \mu \text{ in probability}$$

6.6 6.6 Central Limit Theorem

Theorem 6.6. Let X_i be i.i.d. with mean μ and variance σ^2 . Then

$$\sqrt{n}(\bar{X}_n - \mu) \rightarrow N(0, \sigma^2)$$

7 7. Exercises

Exercise 7.1. Prove that a sigma-algebra is closed under finite unions.

Exercise 7.2. Show that independence implies zero covariance, but not vice versa.

Exercise 7.3. Prove Jensen's inequality for discrete random variables.

Exercise 7.4. Simulate the Law of Large Numbers in R.

```
library(ggplot2)
set.seed(123)
x <- rnorm(1000)
means <- cumsum(x)/seq_along(x)
df <- data.frame(n = seq_along(means), mean = means)
ggplot(df, aes(x = n, y = mean)) +
  geom_line() +
  theme_classic() +
  labs(x = "Sample size", y = "Mean")
```

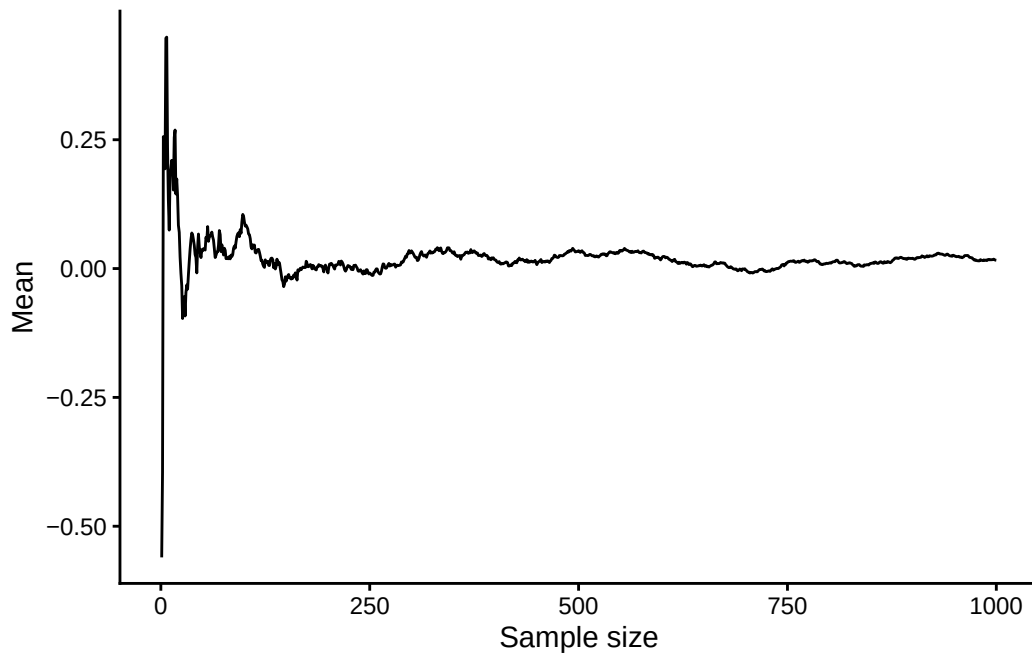


Figure 1: Simulating the Law of Large Numbers

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